Chapter test

26 Basic genetics

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. If 20% of a sample of DNA is adenine, what percentage of the DNA is guanine?

   A. 20%
   B. 30%
   C. 40%
   D. It cannot be determined from the information given. ☐

2. In fruit flies, wings can be long or vestigial. A female fruit fly with vestigial wings was crossed with a male with long wings. All the F_1 offspring produced had long wings. Two of the F_1 offspring were crossed with each other. What would be the probability of the F_2 offspring having vestigial wings?

   A. 25%
   B. 50%
   C. 75%
   D. 100% ☐

3. In mice, fur colour is controlled by a pair of alleles. The allele for black fur is dominant to the allele for brown fur. Which of the following confirms that a mouse with black fur is a heterozygote?

	Cross	***Phenotype of the F_1 offspring***
A.	Cross it a mouse with black fur	All have black fur
B.	Cross it a mouse with black fur	The ratio of black-fur offspring to brown-fur offspring is around 3:1
C.	Cross it a mouse with brown fur	All have black fur
D.	Cross it a mouse with brown fur	The ratio of black-fur offspring to brown-fur offspring is around 1:1 ☐

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★ 4. In guinea pigs, fur length and fur colour are controlled by two different genes. A pure-breed guinea pig with long brown fur was crossed with a pure-breed guinea pig with short black fur. They gave birth to four offspring which had short brown fur.

Which of the following conclusions can be drawn based on the above case?

(1) Short fur is dominant over long fur.
(2) The four offspring have the same genotype for fur length and fur colour.
(3) The genes controlling fur length and fur colour are located on different chromosomes.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

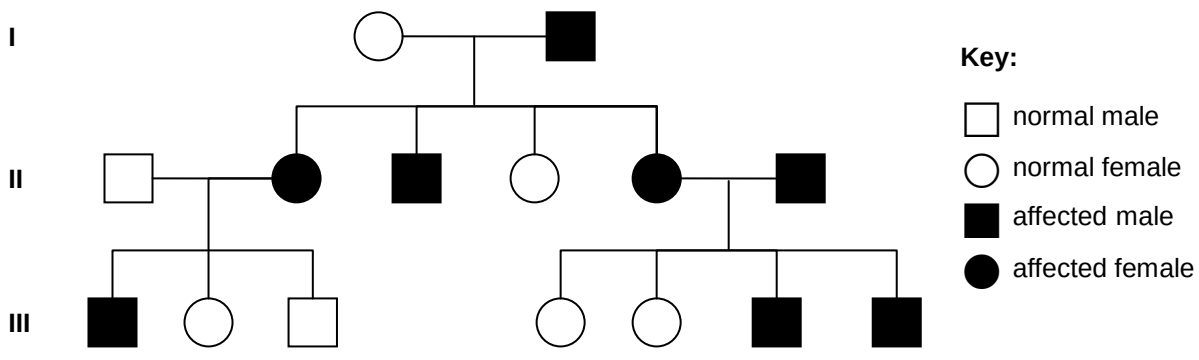
5. A man has blood group A and his wife has blood group AB. What are the possible blood groups of their children?

A. AB only
B. A and B only
C. A and AB only
D. A, B and AB only ☐

6. Cystic fibrosis is an inherited disease caused by a recessive allele. A child is born with cystic fibrosis despite neither parent having the disease. The parents are going to have a second child. What is the chance that the second child will have cystic fibrosis?

A. 100%
B. 75%
C. 50%
D. 25% ☐

★ 7. The pedigree below shows the inheritance of a trait in a family.



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What is the type of inheritance of this trait?

A. autosomal dominant
B. X-linked dominant
C. X-linked recessive
D. Y-linked dominant

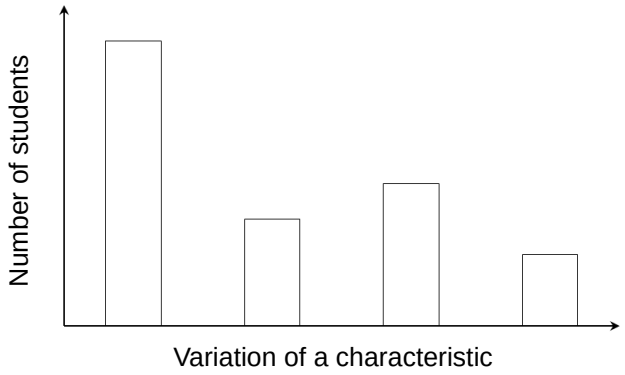
8. Which of the following combinations is correct?

	Sex chromosome in the mother's ovum	*Sex chromosome in the father's sperm*	*Sex of the baby*
A.	X	X	male
B.	X	Y	male
C.	Y	X	female
D.	X	Y	female

9. How is red-green colour blindness inherited?

A. Males inherit the condition from their fathers only.
B. Males can inherit the condition from their mothers having normal colour vision.
C. Females inherit the condition from their fathers only.
D. Females inherit the condition from their mothers only.

10. A certain characteristic of a group of students were measured and the data are presented in the graph below.



Which of the following is most probably the characteristic being studied?

A. hand span
B. intelligence quotient
C. sex
D. blood group

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Question	1	2	3	4	5	6	7	8	9	10
Related section	26.2	26.3	26.3	26.3	26.4	26.4	26.4	26.4	26.4	26.5

B. Problem solving (23 marks)

11. The coat colour of Labrador dogs is controlled by a pair of alleles. The allele for black coat (B) is dominant to the allele for yellow coat (b).

A breeder wants to breed from a black Labrador dog he has recently bought, but he does not know its genotype.

(a) State the meaning of the term 'genotype'. (1 mark)

(b) How could the breeder find out the genotype of the Labrador dog? Describe the cross he could carry out and the expected results. (3 marks)

(c) Complete the Punnett squares below to show the possible results of this cross. (2 marks)

Gametes		

Gametes		

(Total: 6 marks)

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★ 12. The feather colour of a certain breed of chicken is inherited in a very similar way to blood groups in humans. The gene for feather colour has two codominant alleles:

C^B which is the allele for black feather

C^W which is the allele for white feather

The table below shows the genotypes and phenotypes of different feather colours seen in a herd of chickens.

Genotype	Phenotype
$C^B C^B$	black
$C^W C^W$	white
$C^B C^W$	speckled

(a) Explain the term ‘codominance’. (2 marks)

(b) Use a genetic diagram to show the result of a cross between two speckled chickens. (4 marks)

(Total: 6 marks)

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★ 13. The table shows some of the characteristics of four members in a family.

	Mr. Wong	**Mrs. Wong**	**Ken**	**Betty**
Blood group	B	O	B	O
Height	178	165	168	160
Shape of earlobes	Free earlobes	Free earlobes	Attached earlobes	Free earlobes

(a) Deduce, with reasons, the blood group genotype of Mr. Wong.
(Marks will not be awarded to genetic diagrams.) (4 marks)

(b) In humans, the shape of earlobes is determined by a pair of alleles. Based in the given information, deduce whether the allele for free earlobes is dominant or recessive.
(Marks will not be awarded to genetic diagrams.) (5 marks)

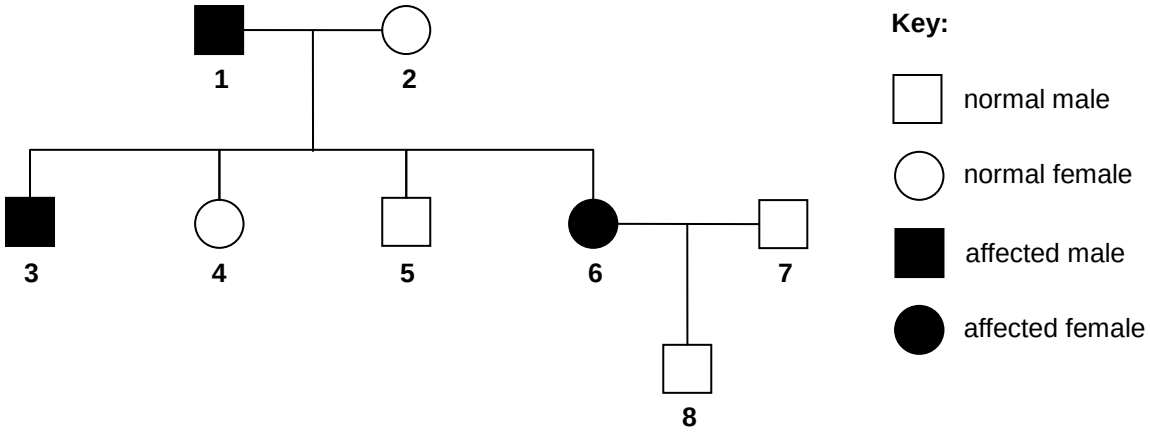
(c) Explain why Ken and Betty are of different sexes despite the fact that they are a pair of twins. (2 marks)

(Total: 11 marks)

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C. Exam challenge (7 marks)

★★ 14. The pedigree below shows the inheritance of a disease in a family. The disease is caused by a dominant allele (A).



(a) Using information in the pedigree, explain why the inheritance of the disease is ***not*** sex-linked. (3 marks)

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(b) Individuals 6 and 7 are going to have a second child. What will be the chance of this child having the disease? Explain your answer by means of a genetic diagram. (4 marks)

(Total: 7 marks)

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Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	B	2.	A	3.	D	4.	A	5.	D
6.	D	7.	A	8.	B	9.	B	10.	D

(1 mark each)

B. Problem solving (23 marks)

11. (a) The genetic makeup of an organism / the alleles that determine a characteristic of an organism (1)

(b) To cross the black Labrador dog with a yellow Labrador dog. (1)
If all the puppies are black then the black Labrador dog is homozygous / BB. (1)
If there are yellow puppies in the litter then the black Labrador dog is heterozygous / Bb (1)

(c) 1 mark for each correct Punnett square (1, 1)

Gametes	B	B
b	Bb	Bb
b	Bb	Bb

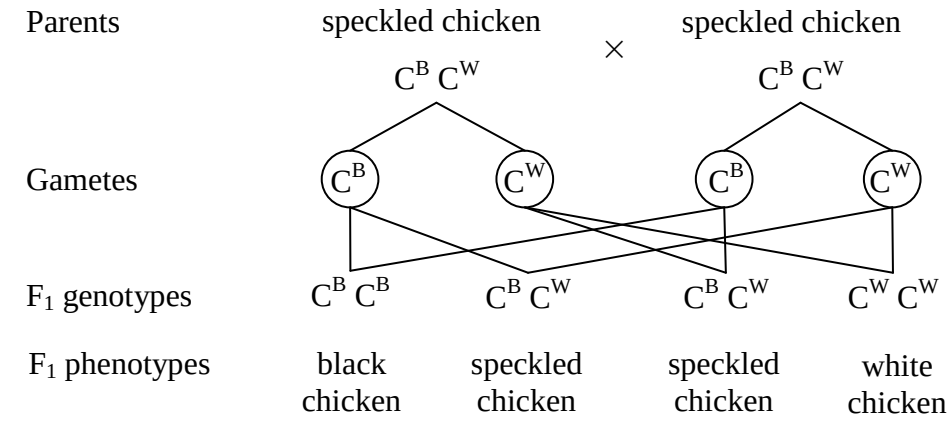
Gametes	B	b
b	Bb	bb
b	Bb	bb

(Total: 6 marks)

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12. (a) It means that both alleles are expressed (1)
in a heterozygote. (1)

(b)



(1)

(1)

(1)

Phenotypic ratio 1 black : 2 speckled : 1 white (1)

(Total: 6 marks)

13. (a) I^Bi (1)
Mr. Wong has blood group B, and so he must have at least one I^B allele. (1)
Betty has blood group O, and so her genotype must be ii. (1)
Betty must have received one i allele from each of her parents. (1)
Therefore, Mr. Wong must have an i allele.

(b) Mr. and Mrs. Wong have free earlobes, and so each of them must carry at least one allele for free earlobes. (1)
Ken has attached earlobes, and so he must have received at least one allele for attached earlobes from his parents. (1)
At least one of the parents is heterozygous. (1)
In a heterozygous condition, only the dominant allele is expressed. (1)
Therefore, the allele for free earlobes is dominant. (1)

(c) The twins are formed from the fusion of two ova with two sperms. (1)
The two sperms carry different sex chromosomes. (1)

(Total: 11 marks)

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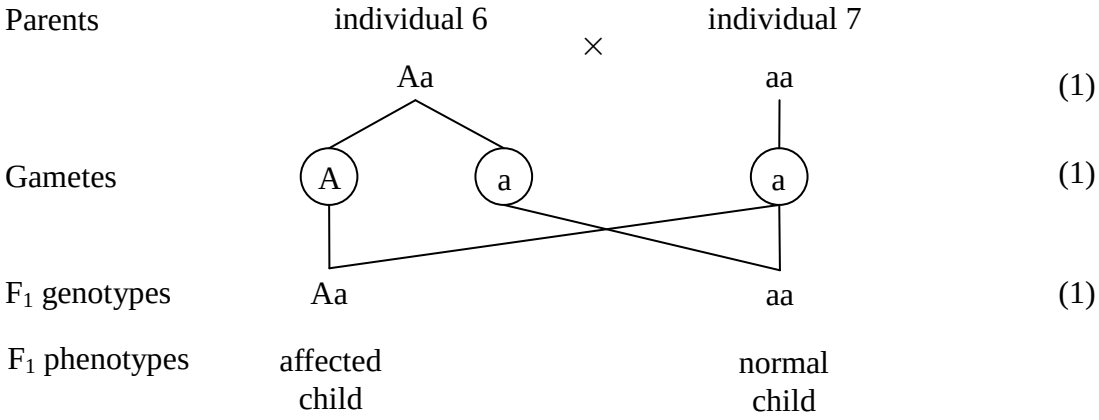
C. Exam challenge (7 marks)

14. (a) The disease is not Y-linked because individual 6 does not have a Y chromosome but still has inherited the disease. (1)

If the disease is X-linked, individual 3 must have inherited the Y chromosome from individual 1 and the X chromosome bearing the normal allele from individual 2 that he would not be affected by the disease. (1)

However, he has the disease and so the disease is not X-linked. (1)

(b)



The chance of the second child having the disease is 1/2 (50%). (1)

(Total: 7 marks)

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